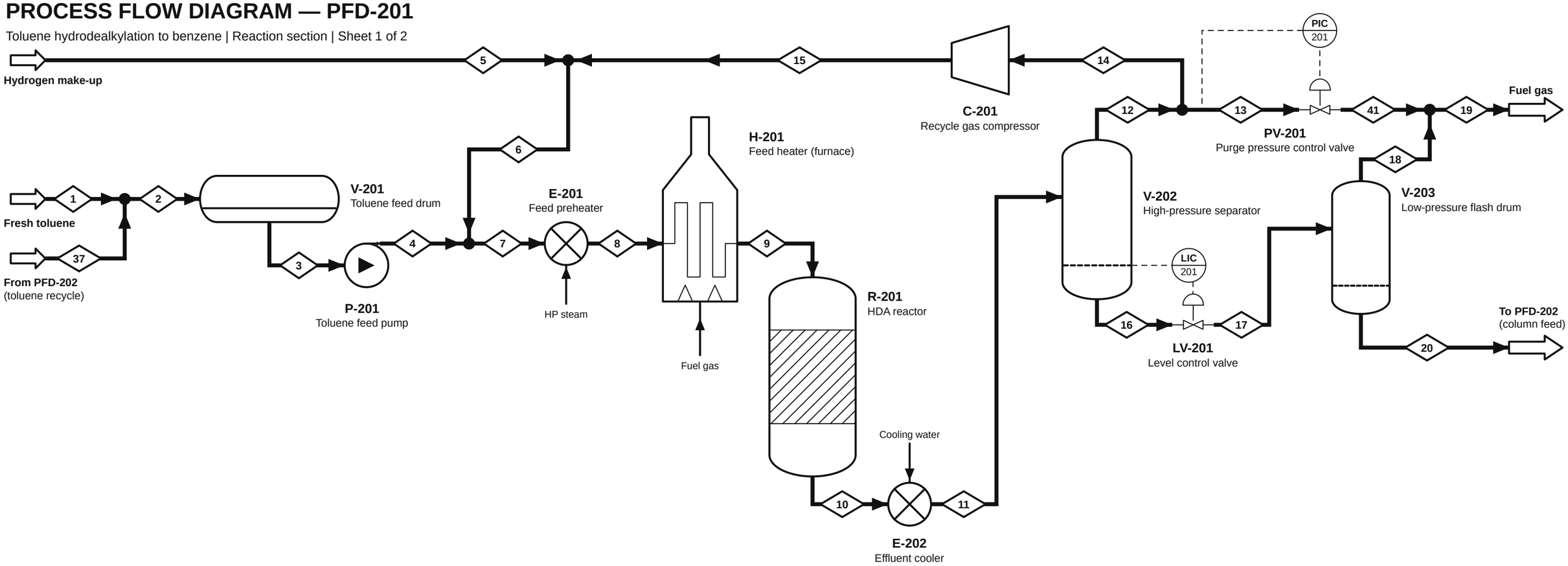
PROCESS FLOW DIAGRAM — PFD-201

Toluene hydrodealkylation to benzene | Reaction section | Sheet 1 of 2



Stream table: sample values (flows from a simple component-balance model; temperatures and pressures nominal, column temperatures near ideal bubble points)

Stream	Phase	Temperature (°C)	Pressure (kPa)	Total (kmol/h)	Hydrogen (kmol/h)	Methane (kmol/h)	Benzene (kmol/h)	Toluene (kmol/h)	Diphenyl (kmol/h)
1	Liquid	25	190	100.0	0.0	0.0	0.0	100.0	0.0
2	Liquid	50	160	132.5	0.0	0.0	0.5	132.0	0.0
3	Liquid	50	160	132.5	0.0	0.0	0.5	132.0	0.0
4	Liquid	51	2700	132.5	0.0	0.0	0.5	132.0	0.0
5	Vapour	25	2600	254.1	241.4	12.7	0.0	0.0	0.0
6	Vapour	40	2550	974.8	661.0	311.4	2.2	0.2	0.0
7	Mixed	45	2550	1107.3	661.0	311.4	2.7	132.2	0.0
8	Vapour	230	2500	1107.3	661.0	311.4	2.7	132.2	0.0
9	Vapour	600	2450	1107.3	661.0	311.4	2.7	132.2	0.0
10	Vapour	650	2400	1107.3	562.3	410.6	100.9	33.0	0.5
11	Mixed	38	2350	1107.3	562.3	410.6	100.9	33.0	0.5
12	Vapour	38	2350	961.1	559.5	398.3	3.0	0.3	0.0
13	Vapour	38	2350	240.4	139.9	99.6	0.8	0.1	0.0
14	Vapour	38	2350	720.7	419.6	298.7	2.2	0.2	0.0
15	Vapour	45	2550	720.7	419.6	298.7	2.2	0.2	0.0
16	Liquid	38	2350	146.2	2.8	12.3	97.9	32.7	0.5
17	Mixed	37	280	146.2	2.8	12.3	97.9	32.7	0.5
18	Vapour	37	280	16.0	2.7	11.1	2.0	0.2	0.0
19	Vapour	37	250	256.4	142.6	110.7	2.8	0.3	0.0
20	Liquid	37	280	130.2	0.1	1.2	95.9	32.5	0.5
37	Liquid	121	350	32.5	0.0	0.0	0.5	32.0	0.0
41	Vapour	35	260	240.4	139.9	99.6	0.8	0.1	0.0

Abbreviations

kmol/h: kilomoles per hour; kPa: kilopascals (absolute); °C: degrees Celsius
HP steam: high-pressure steam
LIC: level indicating controller; LV: level control valve
PIC: pressure indicating controller; PV: pressure control valve
Mixed: vapour-liquid mixture
Stream numbers in diamonds; continuation streams keep their number on both sheets

Toluene hydrodealkylation (HDA) plant — PFD-201	
Reaction section | continued on PFD-202	
Sample process flow diagram; stream values are sample values	
Originator: Claude Code | 2026-10-24	Revision 1 | Sheet 1 of 2